PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GORVEMENT
FORM IV TAHOSSA NORTHWESTERN ZONE MOCK EXAMINATION 2025
032/2B CHEMISTRY 2B (ACTUAL PRACTICAL B)

TIME 2:30HRS **AUGUST 2025**

INSTRUCTIONS

1. This paper consists of two (2) questions. Answer all questions
2. Each question carries twenty-five (25) marks
3. Qualitative Analysis Guide sheet authorized by NECTA and non – programmable calculators may be used.
4. Cellular phones are not allowed in the examination room
5. Write your examination number on every page of your answer booklet(s)
6. You may use the following constant:

Atomic masses:

C = 12, H = 1, Na = 23, S=32, Cl =35.5.

1litre = $1dm^3$ = $1000cm^3$.

Question 1

You are provided with solution GG and HH. One of these solutions is acidic and the other is basic. The basic solution is made by dissolving 7.15g of hydrated sodium carbonate. ($Na_2CO_3.XH_2O$) in distilled water to make up to $500cm^3$ of the solution. The acidic solution was made by dissolving 1.825g of hydrochloric acid (HCl) in distilled water to make $500cm^3$ of the solution.

Procedures:

(i) Pour about $2cm^3$ of solution GG into a test tube and use a litmus paper to test if it is acid or base.

(ii) Discard the content and wash the test tube.

(iii) Repeat the procedure in (i) and (ii) using solution HH

(iv) Titrate the acid solution against the base solution using two or three drops of methyl orange indicator to the end point

(v) Repeat the step (iv) to obtain three more readings and record the result in tabular form.

Question

(a) (i) What was the volume of pipette used?

(ii) What was the colour change during the filtration process?

(iii)Calculate the average volume of the acid used to neutralize the base.

(b) Write the balanced chemical equation for the neutralization reaction between GG and HH

(c) Calculate the concentration of the base in:

(i) Mol/dm^3

(ii) g/dm^3

(d) Determine the value of x in the formula $Na_2CO_3.XH_2O$

Question 2

You are provided with the following

LL: A solution of 0.13M sodium thiosulphate

NN: A solution of 2M HCl

Distilled water

Stop watch or clock

A sheet of white paper marked x.

Procedures:

(i) Measure 25cm^3 of solution LL and put into a 100cm^3 beaker
(ii) Place the beaker containing solution LL on the top of letter X drawn on a sheet of paper.
(iii) Measure 10cm^3 of NN, pour into a beaker containing solution LL and immediately start the stop watch. Swirl the beaker with contents twice.
(iv) Look down vertically through the mouth of the beaker so as to see the cross at the bottom of the beaker. Stop the clock when the letter X is invisible.
(v) Record the time taken for the letter X to disappear completely.
(vi) Repeat the experiment using the data shown in the following table.

Table of experiment

Experiment No.	**Volume of NN (cm^3)**	**Volume of LL in (cm^3)**	**Volume of Distilled water (cm^3)**	**Time (s)**	$^1/_t\ (sec^{-1})$
1	10	25	0		
2	10	20	5		
3	10	15	10		
4	10	10	15		

Questions

(a) Complete by filling the experimental table
(b) What does $^1/_t$ represent in experimental table?
(c) Write the balanced chemical equation for the reaction between LL and NN
(d) How was the factor of concentration varying in this experiment?
(e) Plot a graph of volume of solution LL against $^1/_t$.
(f) Use the graph you have drawn in (e) above to explain how the variation of concentration affects the rate of chemical reaction.